REVIEW

Long-Term Outcomes in Antibody-Negative Autoimmune Encephalitis

A Systematic Review and Meta-Analysis

Prachi Mohapatra,[1] Meena Chandu,[1] Pradeep Kumar,[2] Rajesh Kumar Singh,[1] Deepti Vibha,[1] and Manjari Tripathi[1]

Correspondence
Dr. Vibha
deeptivibha@gmail.com

Neurol Clin Pract. 2026;16:e200602. doi:10.1212/CPJ.0000000000200602

MORE ONLINE

Supplementary Material

Abstract

Purpose of Review

Seronegative autoimmune encephalitis (AE), characterized by the absence of known neuronal autoantibodies despite meeting clinical criteria, presents significant diagnostic and therapeutic challenges. Although immunotherapy is frequently used, long-term outcomes in this population remain poorly characterized. This systematic review and meta-analysis aims to synthesize evidence on long-term outcomes in seronegative AE, focusing on functional status, seizure burden, cognitive impairment, relapses, immunotherapy practices, and prognostic indicators. We included studies involving patients 12 years or older diagnosed with antibody-negative probable AE (ANPRA), possible AE, or definite limbic encephalitis (LE) with negative neuronal antibodies according to Graus et al. (2016), with a minimum median follow-up of 6 months.

Recent Findings

Nineteen studies, predominantly retrospective cohorts with moderate risk of bias, met the inclusion criteria. The pooled proportion of patients achieving favorable functional outcomes (modified Rankin Scale 0–2) was 65% (95% CI 51% – 78%; I^2 = 88.1%; 12 studies, n = 782). Outcomes differed by seronegative subtype, with better functional recovery in seronegative LE (73%; 95% CI 45%–90%; 6 studies, n = 95) compared with ANPRA (47%; 95% CI 38%–55%; 5 studies, n = 277; p = 0.005). When compared with seropositive AE (n = 466), seronegative patients (n = 373) demonstrated comparable odds of favorable functional outcomes (odds ratio 0.50; 95% CI 0.19–1.31; p = 0.16) but had a significantly higher risk of persistent seizures (risk ratio 1.66; 95% CI 1.01–2.72; p = 0.04). The pooled relapse proportion among seronegative patients was 17% (95% CI 11%–26%; 7 studies, n = 350), with no significant difference by serostatus. Residual disability assessed using the Clinical Assessment Scale in Autoimmune Encephalitis suggested mild persistent impairment (mean score 3.11; 95% CI 1.79–4.43; 5 studies, n = 409). Owing to heterogeneity and inconsistent reporting, meta-analysis of cognitive outcomes, immunotherapy strategies, and RAPID (Refractory status epilepticus, Age ≥ 60 years, Probable AE, Infratentorial involvement, Delayed immunotherapy ≥ 1 month) scores was not feasible; narrative synthesis suggested persistent cognitive deficits and lower utilization of second-line immunotherapy in seronegative cohorts.

Summary

Seronegative AE is associated with heterogeneous outcomes, with antibody-negative LE demonstrating better recovery than ANPRA. Subtype-specific variability and elevated seizure burden highlight the need for individualized management, timely diagnosis, structured follow-up, and well-designed prospective research.

Downloaded from https://www.neurology.org by Cardiff University (cfu) / Wales on 14 March 2026

[1]Department of Neurology, All India Institute of Medical Sciences, New Delhi, India; and [2]Clinical Research Unit, All India Institute of Medical Sciences, New Delhi, India.

Copyright © 2026 American Academy of Neurology

Copyright © 2026 American Academy of Neurology. Unauthorized reproduction of this article is prohibited.

Introduction

Autoimmune encephalitis (AE) encompasses a heterogeneous group of neuroinflammatory disorders marked by subacute cognitive decline, seizures, psychiatric symptoms, and movement disorders. Despite advances in identifying pathogenic neuronal autoantibodies, a significant subset of patients remains seronegative, lacking detectable antibodies while still meeting the clinical criteria for possible AE. As per the widely adopted Graus et al. (2016) criteria, these cases may be classified as antibody-negative probable AE (ANPRA) or definite autoimmune limbic encephalitis (LE) without detectable neuronal antibodies, based on clinical features, imaging, and CSF findings after ruling out other causes.1

Seronegative AE presents a clinical challenge due to its heterogeneous manifestations and lack of specific biomarkers. A recent expert commentary highlighted key diagnostic pitfalls, warning that over-reliance on antibody testing can contribute both to overdiagnosis in the setting of false positives and to underdiagnosis when patients fulfilling clinical criteria are antibody-negative.[2] Several studies have shown that seronegative patients often respond well to immunotherapy, with outcomes comparable with seropositive patients.[3-5] A large cohort study from South Korea by Lee et al. highlighted considerable clinical variability and poorer long-term outcomes in seronegative AE patients. The authors also proposed the RAPID score, a prognostic tool based on refractory seizures, 60 years or older, ANPRA subtype, infratentorial MRI lesions, and delayed immunotherapy.[3]

Despite these insights, no prior meta-analysis has systematically synthesized long-term outcome data specifically in seronegative AE or compared functional recovery, seizure control, relapses, and cognitive outcomes across subtypes or against seropositive cohorts. This systematic review and meta-analysis aims to bridge this gap by evaluating long-term outcomes in seronegative AE, emphasizing functional independence, seizure burden, cognitive impairment, relapse risk, and immunotherapy strategies. By critically integrating data from cohort studies worldwide, this study aims to inform clinical decision-making, highlight the diverse outcomes observed in seronegative AE, and identify key gaps that warrant future prospective research.

Methods

This systematic review and meta-analysis adhered to Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) 2020 guidelines and was prospectively registered in PROSPERO (CRD420250651787).

Eligibility Criteria

Studies involving patients 12 years or older diagnosed with antibody-negative probable AE (ANPRA), possible AE, or definite autoimmune LE with negative antineuronal antibodies, as defined by the Graus et al. (2016) criteria, were included. To ensure the evaluation of long-term outcomes, a minimum median follow-up duration of 6 months was required. Eligible study designs included cohort studies, cross-sectional studies, quasiexperimental studies, and case series comprising at least 5 patients. Studies were required to report at least one relevant long-term clinical outcome. Functional status assessed by the modified Rankin Scale (mRS) was designated as the primary outcome of interest. Secondary outcomes included the Clinical Assessment Scale in Autoimmune Encephalitis (CASE) score,[6] seizure burden, residual cognitive impairment, and relapses. In addition, data on immunotherapy regimens and, where available, prognostic indicators such as the RAPID score were extracted. Case reports, narrative reviews, and studies focusing exclusively on seropositive AE were excluded.

Search Strategy and Study Selection

A systematic search was conducted across PubMed, Embase, Scopus, Web of Science, and Google Scholar for studies published between January 2015 and January 2025. The search strategy included combinations of the terms: "autoimmune encephalitis" AND ("seronegative" odds ratio (OR) "antibody negative") AND "long-term outcomes", along with outcome-related keywords such as mRS, CASE, seizure, relapse, cognition, immunotherapy, and treatment. The full search strategy is detailed in the eMethods. Titles, abstracts, and full-text articles were independently screened by 2 reviewers (P.M. and M.C.) using Rayyan. Any discrepancies were resolved through discussion with a third reviewer (D.V.).

Outcomes

The primary outcome was functional status, assessed using the mRS, with a favorable functional outcome defined as an mRS of 0–2. Secondary outcomes included clinical severity measured by the CASE. The CASE is a 9-item clinician-rated severity scale (total score 0–27) assessing seizures, memory dysfunction, psychiatric symptoms, level of consciousness, language impairment, dyskinesia/dystonia, gait instability/ataxia, brainstem dysfunction, and weakness, with higher scores indicating greater disease severity.[6] Secondary outcomes also included long-term seizure burden, defined as the proportion of patients with persistent seizures (ongoing seizures at the last follow-up) or drug-refractory epilepsy (DRE), defined as failure of ≥2 appropriately chosen antiseizure medications without seizure freedom; residual cognitive impairment; relapse proportion, reported as the percentage of patients experiencing at least 1 relapse; details of immunotherapy received (including both acute and maintenance therapies); and prognostic indicators such as the RAPID score.

Data Extraction and Quality Assessment

Data were extracted using a standardized template that included study design, country, follow-up duration, serostatus (seropositive vs seronegative), clinical phenotype, and reported outcomes. Some studies further subdivided

Downloaded from https://www.neurology.org by Cardiff University (cfu) / Wales on 14 March 2026

Copyright © 2026 American Academy of Neurology. Unauthorized reproduction of this article is prohibited.

seronegative AE into ANPRA and antibody-negative definite LE, while others reported outcomes more broadly by serostatus. In such cases, "seronegative" referred to patients fulfilling Graus et al. (2016) criteria (possible AE, ANPRA, or LE) but testing negative for neuronal antibodies.

As all included studies were cohorts, quality and risk of bias were assessed using the Newcastle-Ottawa Scale (NOS), which evaluates 3 domains: selection (maximum 4 stars), comparability (maximum 2 stars), and outcome assessment (maximum 3 stars), with more stars indicating lower risk of bias. Publication bias was assessed using funnel plot asymmetry and Egger regression test, where appropriate. The certainty of evidence for 4 key outcomes: favorable functional status (mRS 0–2), CASE scores, persistent seizures or DRE, and relapse proportions was assessed using the Grading of Recommendations, Assessment, Development, and Evaluation (GRADE) framework. Domains evaluated included risk of bias, inconsistency, indirectness, imprecision, and publication bias. Certainty ratings were categorized as high, moderate, low, or very low. A Summary of Findings table was generated using the GRADE methodology with the aid of GRADEpro software (eTable 1).

Data Synthesis and Statistical Analyses

A descriptive synthesis of study demographics, treatment strategies, and clinical outcomes was initially performed. Where feasible, quantitative meta-analysis was conducted using random-effects models (DerSimonian and Laird method). Between-study heterogeneity was assessed using the I^2 statistic and τ^2. I^2 values of approximately 25%, 50%, and 75% were considered to represent low, moderate, and high heterogeneity, respectively, and τ^2 was reported as the estimate of between-study variance. Proportions were pooled using inverse-variance methods with logit transformation. Comparative outcomes were analyzed using odds ratios (ORs) or risk ratios (RRs), each reported with 95% CIs. The mean follow-up CASE scores were summarized using pooled means and SDs. Prespecified subgroup analyses were conducted based on AE subtype (ANPRA vs LE) and follow-up duration (<12, 12–24, ≥24 months). Meta-regression was performed to evaluate the association of follow-up duration and risk of bias with functional outcomes. Meta-analyses for cognitive outcomes, immunotherapy practices, and RAPID score were not performed due to insufficient or heterogeneous data. All statistical analyses were conducted in R (version 4.5.0) using the meta and metafor packages.

Standard Protocol Approvals, Registrations, and Patient Consents

This study is a systematic review and meta-analysis based on previously published data and did not involve individual patient data; therefore, institutional ethics approval and informed consent were not required.

Data Availability

All data extracted and analyzed during this study, including tabulated results and the complete search strategy, are provided in the results and supplementary materials. The R code used for statistical analysis will be available from the corresponding author on reasonable request. No individual patient-level data were used.

Results

A total of 1,220 records were identified through searches of PubMed, Embase, Scopus, Web of Science, and Google Scholar. After removing 484 duplicates, 736 titles and abstracts were screened. Of these, 56 full-text articles were assessed for eligibility. Following review, 19 studies met the inclusion criteria and were included in the final analysis. The most common reasons for exclusion were insufficient follow-up (<6 months), focus on seropositive AE, or lack of relevant long-term outcomes.

The study selection process is illustrated in the PRISMA 2020 flow diagram (Figure 1).

Of the 19 included studies:

1. Twelve studies reported long-term outcomes using the mRS,
2. Five studies reported CASE scores,
3. Six studies provided data on seizure outcomes,
4. Seven studies reported on residual cognitive dysfunction,
5. Seven studies reported on relapse proportions,
6. Thirteen studies reported immunotherapy details, and
7. Three studies reported on RAPID score.

These outcome domains were analyzed separately where sufficient data were available.

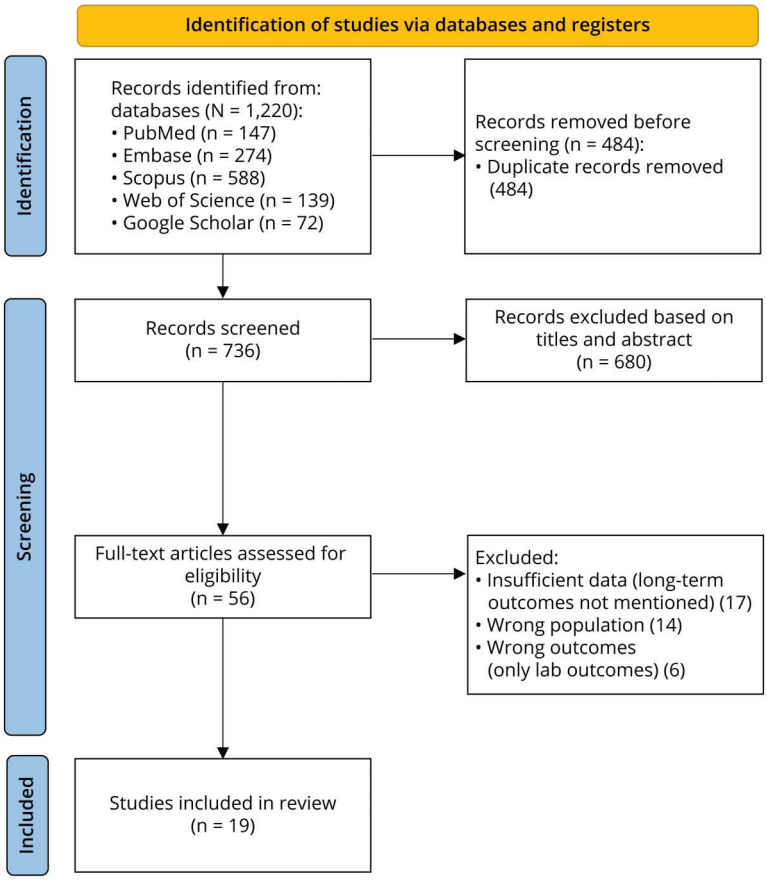
Favorable Functional Outcomes (mRS 0–2) at Long-Term Follow-Up

A total of 12 studies reported long-term functional outcomes in patients with seronegative AE, defined using the mRS dichotomized as favorable (mRS 0–2) and poor (mRS 3–6) outcomes. The characteristics and outcome proportions from these studies are summarized in eTable 2. Across studies, there was considerable variability in sample size, follow-up duration (ranging from 12 to 52.5 months), and outcome proportions, with favorable outcomes reported in 40%–100% of cases. Most studies were retrospective cohorts with moderate risk of bias; 2 cohorts were rated as low risk based on the NOS. Funnel plot inspection did not reveal major asymmetry (eFigure 1). The Egger test for small-study effects was nonsignificant ($t = 1.58$, $df = 10$, $p = 0.15$), indicating no evidence of publication bias. However, residual heterogeneity remained substantial ($\tau^2 = 4.84$), suggesting unexplained between-study variability.

Downloaded from https://www.neurology.org by Cardiff University (cfu) / Wales on 14 March 2026

Copyright © 2026 American Academy of Neurology. Unauthorized reproduction of this article is prohibited.

Figure 1 PRISMA Flow Diagram



A. Pooled Proportion of Favorable Functional Outcomes (mRS 0–2) in Seronegative AE

The pooled proportion of patients achieving favorable long-term functional outcomes (mRS 0–2) across the 12 included studies (n = 782) was 0.65 (95% CI 0.51–0.78), indicating that approximately 65% of patients attained functional independence. However, there was substantial heterogeneity across studies (I^2 = 88.1%, p < 0.0001). The 95% prediction interval ranged from 0.20 to 0.98. The forest plot summarizing these findings is shown in Figure 2. Meta-regression analyses incorporating the risk of bias (low vs moderate) and follow-up duration as moderators did not reveal a significant association with the proportion of patients achieving an mRS score of 0–2 (eTable 3).

B. Subgroup Analyses

By Follow-Up Duration

Subgroup analysis based on follow-up duration was conducted to assess the stability of favorable outcomes over time. All included studies had an overall follow-up duration of at least 12 months. In studies with ≥24 months of follow-up (6 studies, n = 501), the pooled proportion of patients with mRS 0–2 was 0.65 (95% CI 0.44–0.81). By contrast, studies with 12–24 months of follow-up (6 studies, n = 280) showed a pooled proportion of 0.59 (95% CI 0.36–0.78). Despite this numerical difference, the test for subgroup differences was not statistically significant (p = 0.572), indicating no clear difference in favorable outcomes between the 2 follow-up groups (eFigure 2).

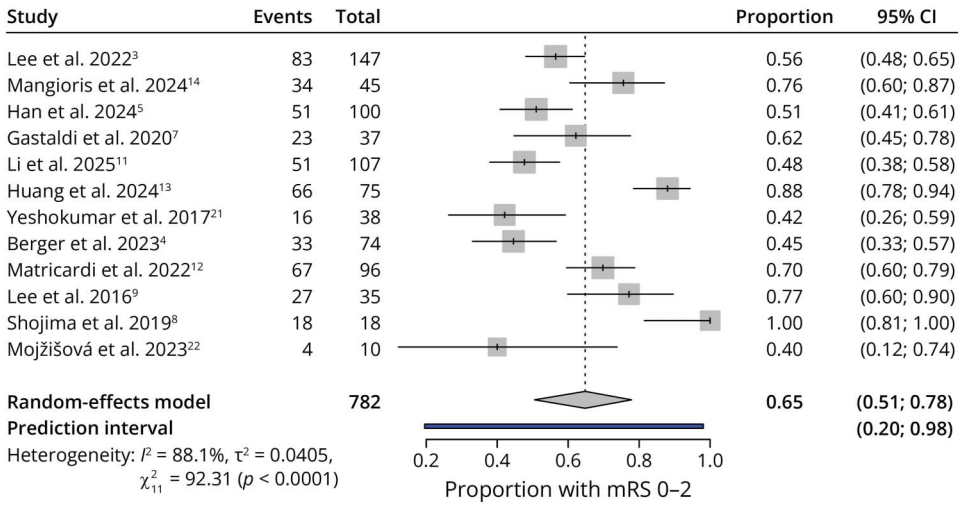
By AE Subtype (ANPRA vs LE)

Favorable outcomes were further analyzed based on AE subtype. Among patients with ANPRA, the pooled proportion of those achieving mRS 0–2 was 0.47 (95% CI 0.41–0.52; 5 studies, n = 277). By contrast, patients with LE demonstrated a higher pooled proportion of favorable outcomes at 0.73 (95% CI 0.45–0.90; 6 studies, n = 95). The difference between subtypes was statistically significant (p = 0.005), favoring better outcomes in the LE subgroup (Figure 3A). In the pooled OR analysis directly comparing ANPRA and LE across matched studies, the odds of achieving favorable outcomes were lower in ANPRA (OR = 0.32, 95% CI 0.09–1.11), although the result did not reach statistical significance. Heterogeneity was moderate (I^2 = 43.6%, p = 0.13), and the 95% prediction interval ranged from 0.03 to 3.85 (Figure 3B).

Downloaded from https://www.neurology.org by Cardiff University (cfu) / Wales on 14 March 2026

Copyright © 2026 American Academy of Neurology. Unauthorized reproduction of this article is prohibited.

Figure 2 Pooled Proportion of Favorable Functional Outcomes (mRS 0–2) in Seronegative Autoimmune Encephalitis



mRS = modified Rankin Scale.

Comparison of Functional Outcomes (mRS 0–2) Between Seronegative and Seropositive AE

Seven studies directly compared favorable long-term functional outcomes (mRS 0–2) between seropositive (n = 466) and seronegative (n = 373) AE. The pooled OR was 0.50 (95% CI 0.19–1.31), indicating a nonsignificant trend favoring the seropositive group. There was substantial heterogeneity across studies (I^2 = 67.8%, p = 0.005), and the prediction interval ranged from 0.08 to 3.08. Although some studies reported significantly better functional outcomes in seropositive patients,[4,7] others found no significant difference between the groups.[8,9] (Figure 4).

CASE Scores at Long-Term Follow-Up

Five studies (n = 409) reported the CASE score at the last follow-up in seronegative patients. For studies reporting follow-up CASE scores as medians with interquartile ranges, we estimated the mean and SD using the method proposed by Wan et al[10].(2014). These derived values were incorporated into the meta-analysis to enable standardized pooling of continuous outcomes. The details of the studies included are summarized in eTable 4. The pooled mean CASE score was 3.11 (95% CI 1.79–4.43), indicating mild residual disability, with substantial heterogeneity (I^2 = 92.9%, p < 0.001). The highest score (4.86) was seen in a cohort with intensive care unit (ICU) patients,[11] while others ranged from 2.0 to 3.0 (Figure 5A). Subgroup analysis showed higher pooled CASE scores in ANPRA (3.51; 95% CI −0.29 to 7.31) than LE (1.33; 95% CI 0.58 to 2.08), with a significant between-group difference (p = 0.015), suggesting greater long-term disability in ANPRA (Figure 5B).

Long-Term Seizure Burden

A total of 6 studies reported long-term seizure outcomes in seronegative AE with a minimum follow-up of 6 months, comprising 277 seronegative AE patients. Although reporting granularity varied, most studies highlighted that a significant subset of patients remain at risk for developing chronic epilepsy or drug-resistant epilepsy (DRE) despite immunotherapy and antiseizure treatment (details in eTable 5).

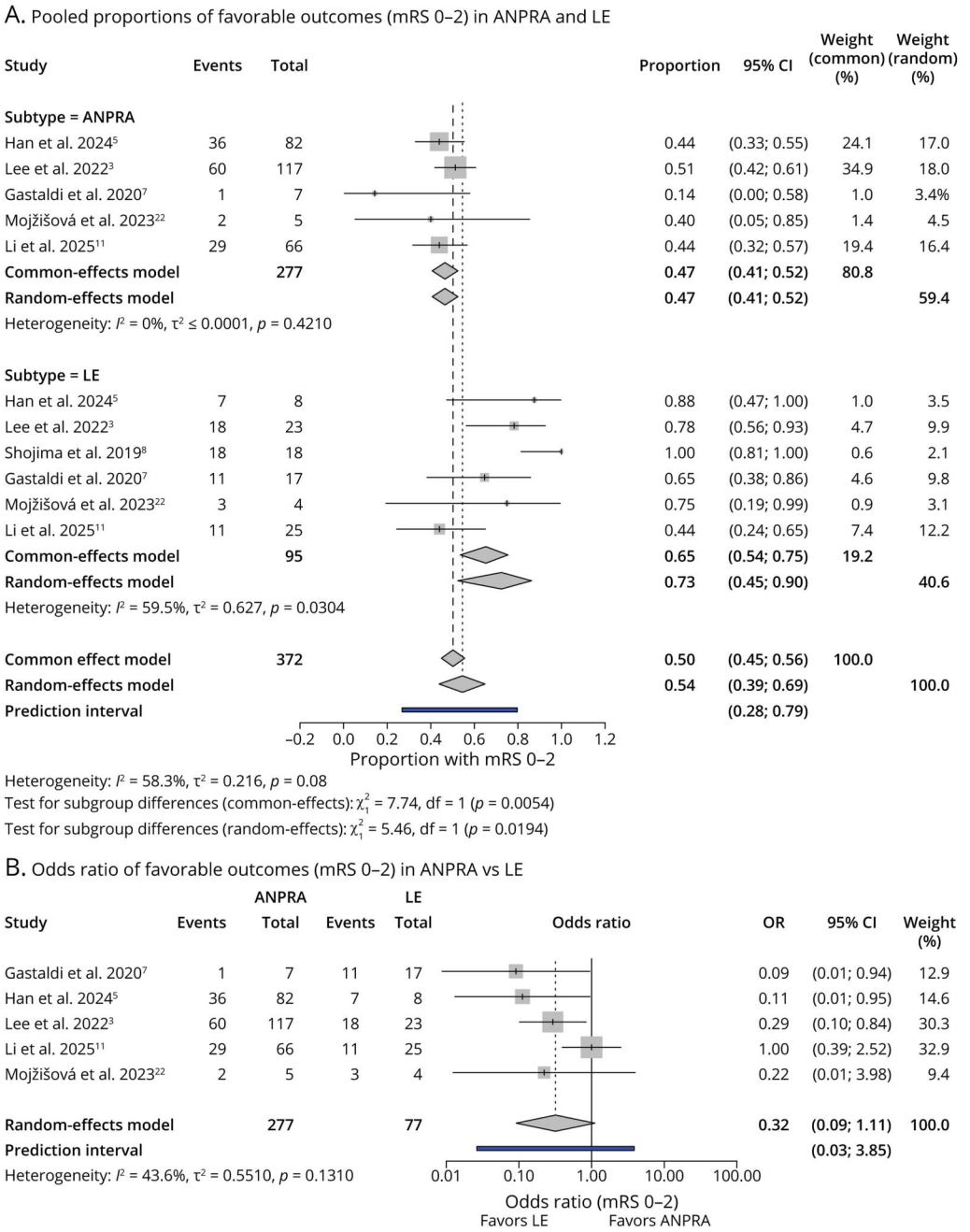
In a large cohort, 58.3% of seronegative vs 35.3% of seropositive patients had persistent seizures at a median follow-up of 34 months; antibody negativity and treatment delay predicted worse outcomes.[12] Another cohort found DRE in 39% of seronegative vs 15% of seropositive patients, with focal seizures, refractory status epilepticus, interictal discharges, and limbic MRI hyperintensities predicting DRE.[13] Rates varied widely across studies (persistent seizures 2.2%–58.3% despite immunotherapy).[12,14-16] A pooled EEG analysis showed greater electroclinical burden in seronegative patients compared with their seropositive counterparts (subclinical seizures 264 vs 23; follow-up EEG seizures 25 vs 23), with multifocal ictal patterns and marked circadian variations more common in seronegative patients.[17]

In a random-effects meta-analysis of 5 studies including patients with seronegative AE (n = 259), the pooled proportion of individuals with persistent seizures or DRE was 38% (95% CI 8%–74%). There was considerable heterogeneity across studies (I^2 = 93.6%, p < 0.0001) (Figure 6A). In a pooled analysis of 3 studies comparing seizure outcomes by antibody status (186 seronegative and 267 seropositive patients), seronegative AE patients had a significantly higher risk of persistent seizures than their seropositive counterparts. The pooled risk ratio (RR) for ongoing seizures in seronegative patients was 1.66 (95% CI 1.01–2.72, p = 0.04). There was no statistical heterogeneity among studies (I^2 = 0%), strengthening the consistency of this finding across cohorts (Figure 6B).

Downloaded from https://www.neurology.org by Cardiff University (cfu) / Wales on 14 March 2026

Copyright © 2026 American Academy of Neurology. Unauthorized reproduction of this article is prohibited.

Figure 3 Subgroup Analysis of Favorable Outcomes (mRS 0–2) by Subtypes (ANPRA vs LE)



(A) Pooled proportions of favorable outcomes (mRS 0–2) in ANPRA and LE. (B) Odds ratio (OR) of favorable outcomes (mRS 0–2) in ANPRA vs LE. ANPRA = antibody-negative probable AE; LE = limbic encephalitis; mRS = modified Rankin Scale.

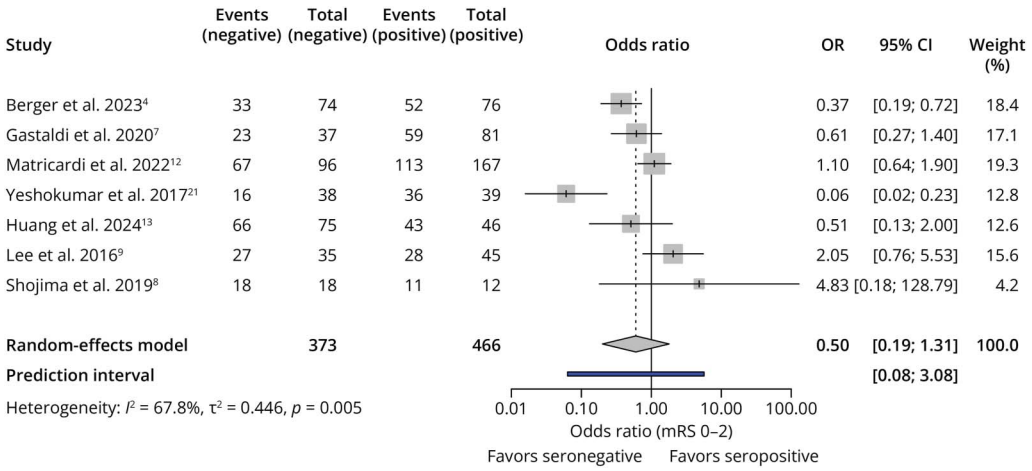
Long-Term Cognitive Outcomes

Cognitive outcomes were reported in 6 studies that included either exclusively seronegative AE patients or separately presented data for seronegative cohorts. Assessment methods varied widely, ranging from bedside tools [e.g., Mini-Mental State Examination (MMSE), Hasegawa Dementia Scale (HDS)] to formal neuropsychological batteries [e.g., Wechsler Adult Intelligence Scale (WAIS), Wechsler Memory Scale (WMS), Wechsler Abbreviated Scale of Intelligence (WASI), California Verbal Learning Test (CVLT), Sydney Language Battery (SydBat), Controlled Oral Word Association Test (COWAT)] and clinical or caregiver reports. Because of this heterogeneity, a formal meta-analysis was not feasible. A systematic narrative synthesis is presented instead. Across studies, the most consistently affected domains were memory, attention, executive function/processing speed, language (including fluency and naming), and visuospatial function. Reported prevalence of impairment varied considerably: for example, memory deficits ranged from 20% to 72%, attention from 13% to 38%,

Downloaded from https://www.neurology.org by Cardiff University (cfu) / Wales on 14 March 2026

Copyright © 2026 American Academy of Neurology. Unauthorized reproduction of this article is prohibited.

Figure 4 Comparison of Functional Outcomes (mRS 0–2) Between Seronegative and Seropositive Autoimmune Encephalitis (AE)



mRS = modified Rankin Scale.

and language-related impairments from 7% to 24% of seronegative patients. The specific assessment methods used and study-level results are summarized in eTable 6.

One study following 45 seronegative AE patients for a median of 36 months found memory impairment in 69%, with additional deficits in attention (38%), visuospatial abilities (22%), orientation (13%), behavior (11%), and language (7%).[14] Another US-based cohort of 15 seronegative patients (median follow-up 14.1 months) found impairments in memory (20%), language (20%), executive function/processing speed (20%), visuospatial ability (13%), apraxia (7%), and intellectual function (7%). In the comparator group of 54 seropositive patients, memory deficits were more common (56%), whereas impairments in other domains were less frequent.[16] The largest cohort reported persistent cognitive deficits in 72% (69/96) of seronegative patients at 24 months, vs 57% (95/167) of seropositive patients.[12] In a study involving patients with definite LE, cognitive testing with MMSE or HDS showed persistent dysfunction in 50% (4/8) of seronegative patients and 83% (5/6) of seropositive patients at 1 year.[8] An Australian cohort assessed 26 seronegative and 24 seropositive patients with WAIS, WMS, WASI, and CVLT at a median of 39 months and found cognitive deficits in both groups, but more pronounced impairments in working, delayed, and auditory memory in seronegative patients; processing speed was reduced in both.[18] A follow-up from the same group assessed 52 AE patients with SydBat, COWAT, and WASI subtests, finding individual-level impairments in 29% (SydBat) and 18% (COWAT). Among seronegative patients, naming (24%) and fluency (20%) impairments were common, with additional deficits on WASI subtests (8%–9%); fluency impairment was significantly more frequent in the seronegative group.[19] Finally, a study of patients with antibody-negative LE demonstrated improved cognitive outcomes in 57% and worsened cognitive outcomes in 32% at a median follow-up of 18 months.[15]

Relapse Outcomes

Relapse proportion, defined as the percentage of patients experiencing at least 1 relapse during follow-up in seronegative AE, was reported in 7 of the included studies (eTable 7). A pooled meta-analysis of 350 patients across these studies yielded a relapse proportion of 17% (95% CI 11%–26%; I^2 = 62.8%, p = 0.013), with a wide prediction interval (5%–47%) (Figure 7A). The highest proportions of relapses were reported in 3 studies, ranging from 18.9% to 31%.[7,14,16] By contrast, 2 studies reported low relapse rates between 0% and 5%.[12,20]

Five studies included comparative data between seronegative and seropositive cohorts (198 seronegative, 363 seropositive patients). A meta-analysis of these revealed no significant difference in relapse risk between the groups, with a pooled OR of 0.87 (95% CI 0.42–1.78, I^2 = 37.9%, p = 0.18) (Figure 7B). While some studies[12,16] observed numerically higher relapse proportions in the seropositive group, others[9] showed higher odds in the seronegative cohort. Notably, one study reported no relapses in either group.[20]

Immunotherapy Details of the Included Studies

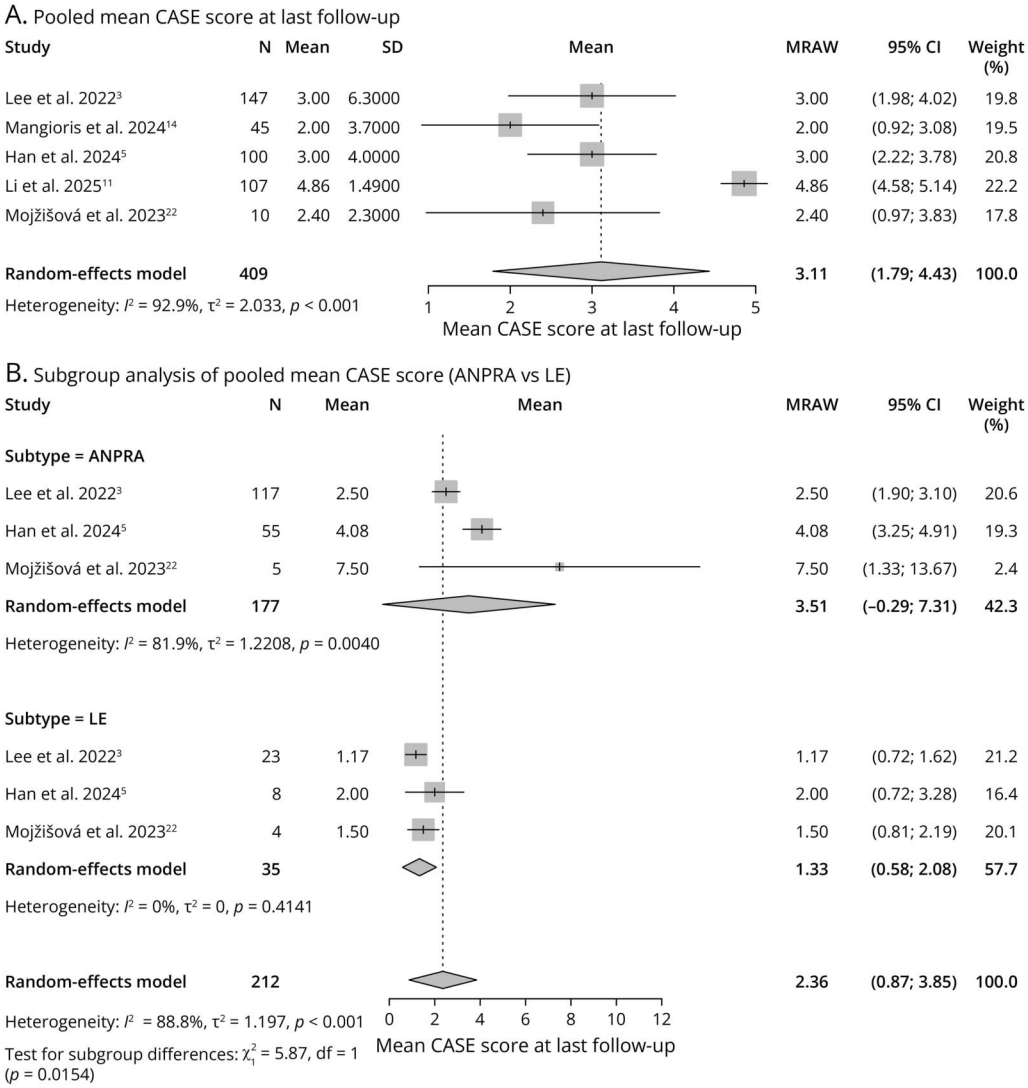
Immunotherapy strategies across the included studies revealed significant variability, particularly in the use of second-line and maintenance agents between seronegative and seropositive AE cohorts. The details are presented in eTable 7.

First-line immunotherapy most frequently included high-dose IV corticosteroids, IV immunoglobulin (IVIG), and plasma exchange (PLEX). Among seronegative AE patients,

Downloaded from https://www.neurology.org by Cardiff University (cfu) / Wales on 14 March 2026

Copyright © 2026 American Academy of Neurology. Unauthorized reproduction of this article is prohibited.

Figure 5 Mean CASE Scores at Last Follow-Up



(A) Pooled mean CASE score at last follow-up. (B) Subgroup analysis of pooled mean CASE score (ANPRA vs LE). ANPRA = antibody-negative probable AE; CASE = Clinical Assessment Scale in Autoimmune Encephalitis; LE = limbic encephalitis.

corticosteroids were used in the majority of cases (ranging from 35.5% to 96%), with IVIG usage varying from 9.7% to 96.6%, and PLEX employed in up to 37.3% cases. Several studies reported a high proportion of patients receiving more than 1 first-line agent, with combination therapy used in 25%–75.5% of patients across cohorts.

In comparison, studies with seropositive cohorts demonstrated broadly similar first-line regimens, although the frequency of IVIG and PLEX use was generally higher in certain studies.[4,16,21] Corticosteroids remained the most universally administered therapy across both groups.

Second-line and maintenance therapies were used more frequently in seropositive patients. For instance, one study reported second-line therapy in 39.5% of seropositive patients vs 13.7% in seronegative patients (p = 0.005).[7] Similarly, another study reported that second-line therapy was administered to 64.7% of seropositive patients compared with 41.7% of seronegative patients, with maintenance therapy prescribed in over 60% of both groups.[12]

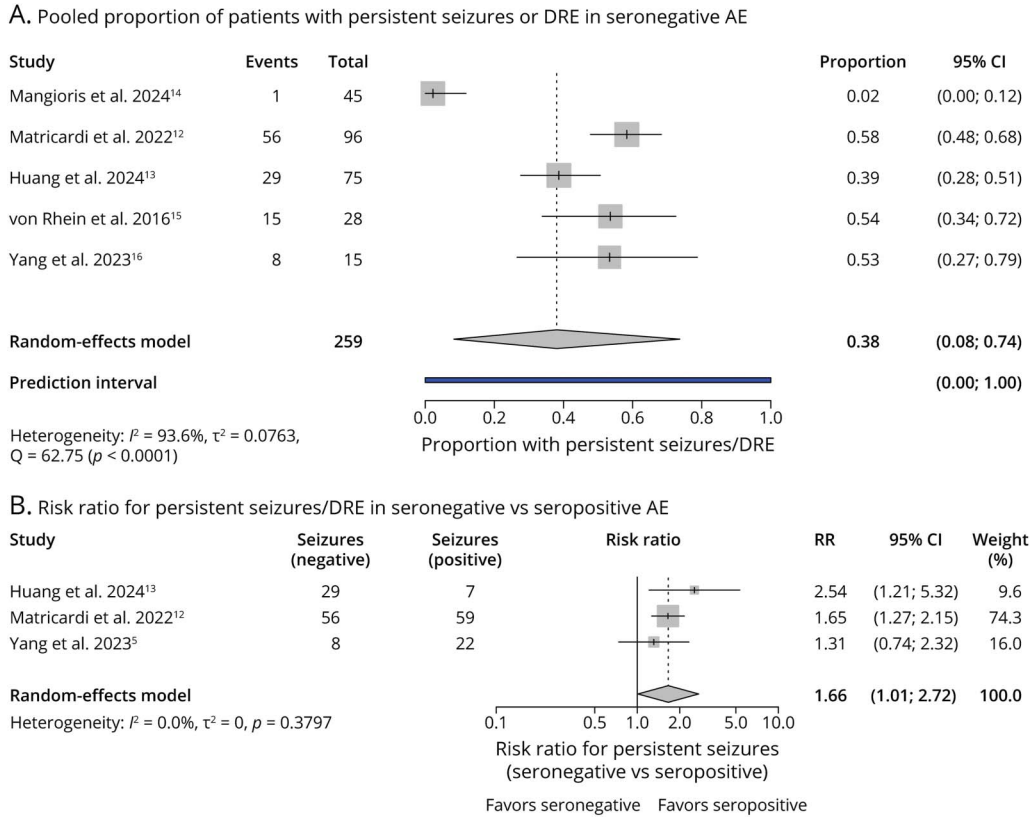
Rituximab was the most commonly used second-line agent across studies, with reported use ranging from 9.7% to 87% among seronegative patients. Other agents including cyclophosphamide, azathioprine, mycophenolate mofetil, and tocilizumab were used with variable frequency across cohorts.

Notably, one study evaluating patients with exclusively seronegative LE reported that immunotherapy could be successfully discontinued in 75% of cases during follow-up, highlighting the potential for sustained remission in carefully selected individuals.[15]

Downloaded from https://www.neurology.org by Cardiff University (cfu) / Wales on 14 March 2026

Copyright © 2026 American Academy of Neurology. Unauthorized reproduction of this article is prohibited.

Figure 6 Pooled Proportion of Persistent Seizures or Drug-Resistant Epilepsy in Seronegative AE and Comparison With Seropositive Cases



(A) Pooled proportion of patients with persistent seizures or DRE in seronegative AE. (B) Risk ratio for persistent seizures/DRE in seronegative vs seropositive AE. AE = autoimmune encephalitis; DRE = drug-resistant epilepsy.

RAPID Scores

Three studies reported data on the RAPID score (Refractory status epilepticus, age ≥60 years, probable AE [ANPRA], infratentorial involvement, delay in immunotherapy ≥1 month), a five-point prognostic tool proposed by Lee et al. (2022) to predict long-term outcomes in seronegative AE.[3] The original study showed a correlation between higher RAPID scores and poorer 2-year outcomes. One external validation study in an independent Chinese cohort reported an area under the curve (AUC) of 0.74 (95% CI 0.64–0.84) for predicting mRS ≥3.[5] Another study involving a Central European cohort provided individual-level RAPID scores; ANPRA/Acute Disseminated Encephalomyelitis (ADEM) patients had a median score of 3 (range: 2–5), and seronegative LE patients had a median score of 2 (range: 1–3). All patients with mRS ≥3 had RAPID scores ≥3, while those with lower scores generally had favorable outcomes.[22]

Discussion

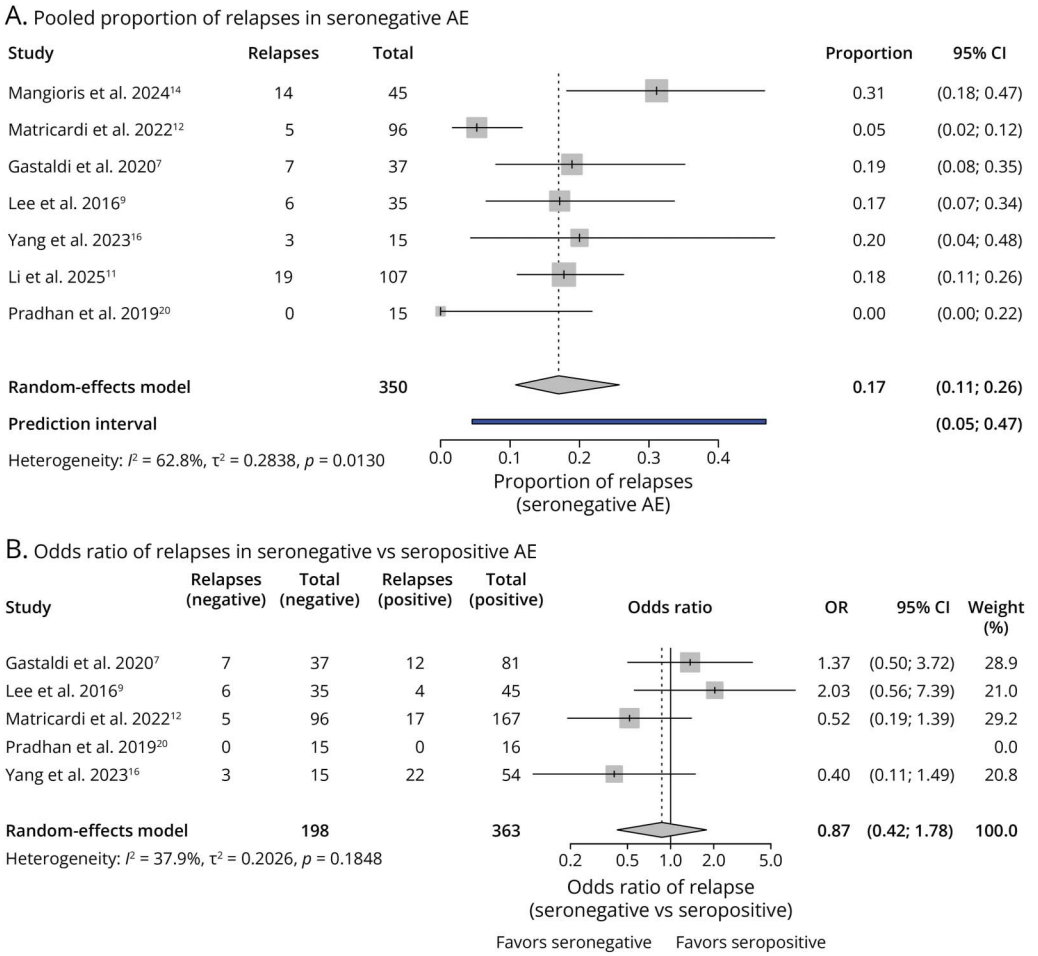
The present systematic review and meta-analysis comprehensively synthesized the available evidence on long-term outcomes in seronegative AE, focusing on functional independence, seizure control, cognitive impairment, relapse rates, and immunotherapy strategies.

Our analysis revealed substantial heterogeneity across the included studies, reflecting the clinical complexity and variability inherent in seronegative AE. On average, approximately 65% of patients achieved favorable long-term functional outcomes (mRS 0–2); however, reported rates ranged widely (40%–100%). This variation likely stems from differences in baseline disease severity, diagnostic delays, and therapeutic approaches. The high level of statistical heterogeneity in the pooled analysis (I^2 > 75%) suggests that the variability in outcomes was driven by true differences across study cohorts rather than random error. Accordingly, the pooled estimate should be interpreted cautiously and should not be directly extrapolated to individual patient care without consideration of clinical context. Interestingly, patients with seronegative LE tended to fare better than those with ANPRA, suggesting meaningful prognostic differences even within seronegative subtypes. These findings highlight the importance of recognizing subtype-specific disease trajectories and support the need for tailored treatment strategies. Although individual studies suggested better recovery in seropositive patients, our pooled analysis did not show

Downloaded from https://www.neurology.org by Cardiff University (cfu) / Wales on 14 March 2026

Copyright © 2026 American Academy of Neurology. Unauthorized reproduction of this article is prohibited.

Figure 7 Pooled Proportion of Relapses in Seronegative AE and OR Compared With Seropositive Cases



(A) Pooled proportion of relapse in seronegative AE. (B) Odds ratio of relapses in seronegative vs seropositive AE. AE = autoimmune encephalitis; OR = odds ratio.

a statistically significant difference, possibly due to heterogeneity in antibody profiles, clinical presentations, and treatment responsiveness across cohorts.

Some of the included studies also enrolled patients with ADEM as part of the broader seronegative AE cohort. While ADEM is traditionally considered a distinct entity, its clinical and immunologic overlap with seronegative AE, especially in adult presentations, makes its inclusion reasonable in the context of long-term outcome studies. To address any potential confounding, we also performed separate subgroup analyses focused on ANPRA and LE, allowing for a more nuanced interpretation of outcomes across different clinical subtypes.

Assessment of long-term clinical severity using the CASE score revealed mild residual disability in seronegative AE, with a pooled mean score of 3.11. Substantial heterogeneity across studies likely reflected differences in baseline severity, follow-up duration, and clinical settings. Higher scores were observed in ICU-based cohorts, emphasizing the impact of disease severity on long-term outcomes. Subgroup analysis showed significantly greater residual disability in ANPRA compared with LE (p = 0.015), reiterating the prognostic value of clinical subtypes.

Persistent seizures emerged as a common and challenging long-term complication, with seronegative AE patients demonstrating a higher risk of chronic epilepsy and DRE compared with their seropositive counterparts. Our analysis showed that seronegative patients were significantly more likely to experience ongoing seizures, with a 66% higher risk compared to those who were seropositive (RR 1.66; 95% CI 1.01–2.72). This suggests that antibody negativity may represent an independent risk factor for long-term seizure burden and potentially drug-resistant epilepsy. Lower exposure to second-line immunotherapies in seronegative cohorts may also have contributed to seizure persistence, although this remains speculative given inconsistent reporting. These findings highlight the need for a more aggressive approach to seizure management in seronegative cases, who may be at greater risk of developing chronic, treatment-resistant epilepsy.

Cognitive impairment was prevalent and sustained, affecting multiple domains including memory, attention, language,

Downloaded from https://www.neurology.org by Cardiff University (cfu) / Wales on 14 March 2026

Copyright © 2026 American Academy of Neurology. Unauthorized reproduction of this article is prohibited.

TAKE-HOME POINTS

- Functional recovery is possible but variable: Approximately 65% of seronegative autoimmune encephalitis (AE), patients achieve favorable long-term functional outcomes (mRS 0–2), although outcomes are significantly better in seronegative limbic encephalitis (LE) than in antibody-negative probable AE (ANPRA).
- Higher long-term seizure burden: Persistent seizures or drug-resistant epilepsy occur in approximately 38% of seronegative AE patients, with a 66% higher risk compared to seropositive cases.
- Cognitive impairment is common and persistent: Residual deficits across memory, language, and executive function are frequent, warranting long-term cognitive monitoring and rehabilitation.
- Immunotherapy is effective but underutilized: Early initiation and escalation of immunotherapy improve outcomes, yet second-line therapies are less frequently used in seronegative AE.
- Relapse risk and prognosis: Relapses occur in approximately 17% of seronegative AE patients, at rates comparable with seropositive cases. Tools such as the RAPID score can help risk-stratify patients and guide the intensity and duration of immunosuppressive therapy.

and executive function, with considerable variability across studies. While formal meta-analysis was limited by methodological heterogeneity, available data suggest a significant and lasting cognitive burden in seronegative AE patients, necessitating long-term cognitive rehabilitation and support.

The observed relapse proportion of 17% highlights a critical need for sustained clinical vigilance and possibly prolonged immunosuppressive strategies in selected high-risk patients, particularly those with recurrent relapses, refractory seizures, or severe initial presentations, as suggested by individual studies.[3,5] However, interpretation of relapse outcomes is limited by substantial heterogeneity across studies, including differences in relapse definitions (clinical events alone vs clinical plus MRI confirmation), follow-up duration, and reporting practices. Interestingly, relapse proportions were comparable between seronegative and seropositive groups, suggesting that shared underlying mechanisms may contribute to relapse, regardless of antibody status.

Immunotherapy practices varied considerably across studies. Corticosteroids, IV immunoglobulin (IVIG), and PLEX constituted the standard first-line treatments for both seronegative and seropositive AE. However, second-line therapies were less consistently employed in seronegative patients, likely reflecting clinical uncertainty regarding treatment efficacy and prognosis in the absence of a defined antibody target. Rituximab was the most commonly used second-line agent among seronegative patients, although its role in this subgroup remains to be clarified through prospective studies. Overall, while first-line immunotherapy was comparable across serostatus, seropositive AE patients were more likely to receive escalated treatment, including second-line and long-term maintenance regimens.

The development of the RAPID score has advanced the understanding of seronegative AE by providing a practical tool to predict long-term outcomes.[3] Its prognostic utility has been externally validated in independent cohorts, demonstrating good accuracy for identifying patients at risk of poor functional recovery.[5,22] Along with evidence that early disability and gait dependence predict poorer outcomes,[14] these findings support RAPID as a simple framework for early risk stratification in seronegative AE patients.

The strengths of this systematic review and meta-analysis are a comprehensive evaluation of most long-term outcomes in seronegative AE, addressing an area that has been relatively underexplored in the existing literature. The inclusion of various outcomes and subgroup analyses offers a clearer understanding of prognostic variability in seronegative autoimmune conditions. However, several limitations must be acknowledged. All included studies were observational, predominantly cohort designs, which restricts the ability to draw causal inferences. Considerable heterogeneity was observed across studies, which may limit the generalizability of the pooled findings. In line with this, the certainty of evidence for key outcomes of functional status, persistent seizures, CASE scores, and relapse proportions was rated as low using the GRADE framework. Moreover, individual participant data were not available, limiting our ability to perform detailed subgroup analyses or adjust for confounding factors.

Conclusion

In conclusion, this systematic review and meta-analysis show that long-term outcomes in seronegative AE are heterogeneous. While many patients achieve functional recovery, persistent seizures and cognitive deficits are frequent long-term complications. Outcomes were generally more favorable in seronegative definite LE than in ANPRA. These findings highlight the importance of individualized management tailored to subtype and emphasize the need for standardized outcome measures. Furthermore, the lack of randomized controlled trials underlines the urgent need for well-designed prospective studies to guide evidence-based care in this challenging subgroup.

Downloaded from https://www.neurology.org by Cardiff University (cfu) / Wales on 14 March 2026

Copyright © 2026 American Academy of Neurology. Unauthorized reproduction of this article is prohibited.

Acknowledgment

The authors thank the investigators of the included studies for making their data publicly available.

Author Contributions

P. Mohapatra: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data; study concept or design; analysis or interpretation of data. M. Chandu: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data; analysis or interpretation of data. P. Kumar: analysis or interpretation of data. R.K. Singh: drafting/revision of the manuscript for content, including medical writing for content; study concept or design. D. Vibha: drafting/revision of the manuscript for content, including medical writing for content; study concept or design; analysis or interpretation of data. M. Tripathi: drafting/revision of the manuscript for content, including medical writing for content.

Study Funding

The authors report no targeted funding.

Disclosure

The authors report no relevant disclosures. Full disclosure form information provided by the authors is available with the full text of this article at Neurology.org/cp.

Publication History

Received by *Neurology® Clinical Practice* July 22, 2025. Accepted in final form December 23, 2025. Submitted and externally peer-reviewed. The handling editor was Deputy Editor Kathryn Kvam, MD.

References

1. Graus F, Titulaer MJ, Balu R, et al. A clinical approach to diagnosis of autoimmune encephalitis. *Lancet Neurol.* 2016;15(4):391-404. doi:10.1016/S1474-4422(15)00401-9
2. Dalmau J, Graus F. Diagnostic criteria for autoimmune encephalitis: utility and pitfalls for antibody-negative disease. *Lancet Neurol.* 2023;22(6):529-540. doi:10.1016/S1474-4422(23)00083-2
3. Lee WJ, Lee HS, Kim DY, et al. Seronegative autoimmune encephalitis: clinical characteristics and factors associated with outcomes. *Brain.* 2022;145(10):3509-3521. doi:10.1093/brain/awac166
4. Berger B, Hauck S, Runge K, Tebartz van Elst L, Rauer S, Endres D. Therapy response in seronegative versus seropositive autoimmune encephalitis. *Front Immunol.* 2023;14:1196110. doi:10.3389/fimmu.2023.1196110
5. Han B, Dai Y, Peng J, Yuan T, Yin Q, Yang L. Study on clinical features and factors related to long-term outcomes of antibody-negative autoimmune encephalitis. *Ann Clin Transl Neurol.* 2024;11(5):1325-1337. doi:10.1002/acn3.52049
6. Lim JA, Lee ST, Moon J, et al. Development of the clinical assessment scale in autoimmune encephalitis. *Ann Neurol.* 2019;85(3):352-358. doi:10.1002/ana.25421
7. Gastaldi M, Mariotto S, Giannoccaro MP, et al. Subgroup comparison according to clinical phenotype and serostatus in autoimmune encephalitis: a multicenter retrospective study. *Eur J Neurol.* 2020;27(4):633-643. doi:10.1111/ene.14139
8. Shojima Y, Nishioka K, Watanabe M, et al. Clinical characterization of definite autoimmune limbic encephalitis: a 30-case series. *Intern Med.* 2019;58(23):3369-3378. doi:10.2169/internalmedicine.3029-19
9. Lee WJ, Lee ST, Byun JI, et al. Rituximab treatment for autoimmune limbic encephalitis in an institutional cohort. *Neurology.* 2016;86(18):1683-1691. doi:10.1212/WNL.0000000000002635
10. Wan X, Wang W, Liu J, Tong T. Estimating the sample mean and standard deviation from the sample size, median, range and/or interquartile range. *BMC Med Res Methodol.* 2014;14(1):135. doi:10.1186/1471-2288-14-135
11. Li Z, He X, Li D, et al. Clinical features and factors associated with outcomes of antibody-negative autoimmune encephalitis in patients requiring intensive care. *Crit Care.* 2025;29(1):24. doi:10.1186/s13054-024-05233-2
12. Matricardi S, Casciato S, Bozzetti S, et al. Epileptic phenotypes in autoimmune encephalitis: from acute symptomatic seizures to autoimmune-associated epilepsy. *J Neurol Neurosurg Psychiatry.* 2022;93(11):1194-1201. doi:10.1136/jnnp-2022-329195
13. Huang W, Zhang H, Li X, et al. Prognostic factors underlying the development of drug-resistant epilepsy in patients with autoimmune encephalitis: a retrospective cohort study. *J Neurol.* 2024;271(8):5046-5054. doi:10.1007/s00415-024-12432-y
14. Mangioris G, Orozco E, Dubey D, et al. Long-term outcomes in antibody-negative autoimmune encephalitis: a retrospective study. *J Neurol.* 2024;271(12):7502-7515. doi:10.1007/s00415-024-12680-y
15. von Rhein B, Wagner J, Widman G, Malter MP, Elger CE, Helmstaedter C. Suspected antibody negative autoimmune limbic encephalitis: outcome of immunotherapy. *Acta Neurol Scand.* 2017;135(1):134-141. doi:10.1111/ane.12575
16. Yang JH, Liu EN, Nguyen L, Dunn-Pirio A, Graves JS. Survival analysis of immunotherapy effects on relapse rate in pediatric and adult autoimmune encephalitis. *Neurology.* 2023;101(22):E2300-E2313. doi:10.1212/WNL.0000000000207746
17. Morano A, Fanella M, Cerulli Irelli E, et al. Seizures in autoimmune encephalitis: findings from an EEG pooled analysis. *Seizure.* 2020;83:160-168. doi:10.1016/j.seizure.2020.10.019
18. Griffith SP, Wesselingh R, Seery N, et al. Characterizing cognitive function in patients with autoimmune encephalitis: an Australian prospective study. *J Neurol.* 2024;271(1):310-324. doi:10.1007/s00415-023-11967-w
19. Griffith SP, Wesselingh R, D'Aprano F, et al. Language impairments in seropositive and seronegative autoimmune encephalitis. *Neurol Sci.* 2024;45(7):3389-3398. doi:10.1007/s10072-024-07382-2
20. Pradhan S, Das A, Das A, Mulmuley M. Antibody negative autoimmune encephalitis-does it differ from definite one? *Ann Indian Acad Neurol.* 2019;22(4):401-408. doi:10.4103/aian.AIAN_206_19
21. Graus F, Escudero D, Oleaga L, et al. Syndrome and outcome of antibody-negative limbic encephalitis. *Eur J Neurol.* 2018;25(8):1011-1016. doi:10.1111/ene.13661
22. Mojžišová H, Krýsl D, Hanzalová J, et al. Antibody-negative autoimmune encephalitis: a single-center retrospective analysis. *Neurol Neuroimmunol Neuroinflamm.* 2023;10(6):e200170. doi:10.1212/NXI.0000000000200170

How to cite this article: Mohapatra P, Chandu M, Kumar P, et al. Long-term outcomes in antibody-negative autoimmune encephalitis: a systematic review and meta-analysis. *Neurol Clin Pract.* 2026;16(2):e200602. doi: 10.1212/CPJ.0000000000200602

Downloaded from https://www.neurology.org by Cardiff University (cfu) / Wales on 14 March 2026

Copyright © 2026 American Academy of Neurology. Unauthorized reproduction of this article is prohibited.